Property 1: Logarithm Reciprocal Rule 🔁📐

Prove: $\log_a b \times \log_b a = 1$

Let $\log_a b = x \Rightarrow a^x = b$ 📝

Let $\log_b a = y \Rightarrow b^y = a$ 📝

Substitute b into the second equation: $(a^x)^y = a$ 🔁

$a^{xy} = a^1$ ⚡

$x \times y = 1$ ✅

Since $x = \log_a b$ and $y = \log_b a$, then $\log_a b \times \log_b a = 1$ 🎯

This also implies: $\frac{1}{\log_b a} = \log_a b$ ➗

Property 2: Product Rule (via Exponents) ➕📊

Prove: $a^{\log_a m + \log_a n} = mn$

Let $\log_a m = x \Rightarrow a^x = m$ 📝

Let $\log_a n = y \Rightarrow a^y = n$ 📝

$a^{\log_a m} \times a^{\log_a n} = a^x \times a^y = a^{x+y}$ ✖️

Since $a^x \times a^y = m \times n$, then $a^{x+y} = mn$ 🧮

Therefore, $a^{\log_a m + \log_a n} = mn$ 🏆

Property 3: Quotient Rule (via Exponents) ➖📉

Prove: $a^{\log_a m - \log_a n} = \frac{m}{n}$

Let $\log_a m = x \Rightarrow a^x = m$ 📝

Let $\log_a n = y \Rightarrow a^y = n$ 📝

$a^{\log_a m - \log_a n} = a^{x-y}$ ⚡

$a^{x-y} = \frac{a^x}{a^y} = \frac{m}{n}$ ➗

Therefore, $a^{\log_a m - \log_a n} = \frac{m}{n}$ ✅

Property 4: Addition of Bases ➕🔢

Prove: $a^{\log_a m} + a^{\log_a n} = m + n$

Let $\log_a m = x \Rightarrow a^x = m$ 📝

Let $\log_a n = y \Rightarrow a^y = n$ 📝

Substitute the exponential forms: $a^x + a^y = m + n$ ✔️

Property 5: Subtraction of Bases ➖🔢

Prove: $a^{\log_a m} - a^{\log_a n} = m - n$

Let $\log_a m = x \Rightarrow a^x = m$ 📝

Let $\log_a n = y \Rightarrow a^y = n$ 📝

Substitute the exponential forms: $a^x - a^y = m - n$ ✔️

Learn (Basic Identity) 🧠✨

$\log_a a^x = x \log_a a = x \times 1 = x$ 🎯

Property 6: Power Rule (via Exponents) ⚡📈

Prove: $a^{n \log_a m} = m^n$

Let $\log_a m = q \Rightarrow a^q = m$ 📝

$a^{n \log_a m} = a^{nq}$ 🔁

Using exponent rules: $a^{nq} = (a^q)^n$ 🧮

Substitute m: $(m)^n = m^n$ 🏆

Property 7: Base Exchange Property (Change of Base) 🔁🔀

Method 1 (Your Method - Exponents): 🧠🧬

Prove: $\log_b a = \frac{\log_x a}{\log_x b}$

Let $K = \log_b a \Rightarrow b^K = a$ 📝

Let $L = \log_x a \Rightarrow x^L = a$ 📝

Let $P = \log_x b \Rightarrow x^P = b$ 📝

Substitute b into the first equation: $(x^P)^K = a \Rightarrow x^{PK} = a$ 🔄

Compare with $x^L = a$: $x^{PK} = x^L$ ⚖️

$PK = L \Rightarrow K = \frac{L}{P}$ ➗

Substitute original variables back: $\log_b a = \frac{\log_x a}{\log_x b}$ ✅

Method 2 (Alternative Method - Power Rule): 🏎️💨

Start with the fundamental identity: $a = b^{\log_b a}$ 🎯

Take $\log_x$ on both sides: $\log_x(a) = \log_x\left(b^{\log_b a}\right)$ ⚖️

Apply the Power Rule: $\log_x(a) = (\log_b a) \times \log_x(b)$ ⚡

Divide by $\log_x(b)$: $\frac{\log_x(a)}{\log_x(b)} = \log_b a$ 🏆

Property 8: Power on Base and Argument 💥🔥

Prove: $\log_{a^n}(x^m) = \frac{m}{n}\log_a(x)$

Method 1 (Your Method - First Principles / Exponents): 🧬📝

Let $y = \log_{a^n}(x^m)$ 📝

Convert to exponential form: $(a^n)^y = x^m$ 🔄

Multiply exponents: $a^{ny} = x^m$ ✖️

Take the m-th root (raise both sides to $\frac{1}{m}$): $(a^{ny})^{\frac{1}{m}} = (x^m)^{\frac{1}{m}}$ ➗

$a^{\frac{ny}{m}} = x$ ⚡

Convert back to log with base a: $\log_a(x) = \frac{ny}{m}$ 🔄

Solve for y: $y = \frac{m}{n}\log_a(x)$ 🧮

Substitute y back: $\log_{a^n}(x^m) = \frac{m}{n}\log_a(x)$ ✅

Method 2 (Alternative Method - Change of Base): 🏎️💨

Start with expression: $\log_{a^n}(x^m)$ 🎯

Apply Change of Base to base a: $\frac{\log_a(x^m)}{\log_a(a^n)}$ 🔀

Apply Power Rule to numerator and denominator: $\frac{m \cdot \log_a(x)}{n \cdot \log_a(a)}$ ⚡

Since $\log_a(a) = 1$: $\frac{m \cdot \log_a(x)}{n \cdot 1} = \frac{m}{n}\log_a(x)$ 🏆